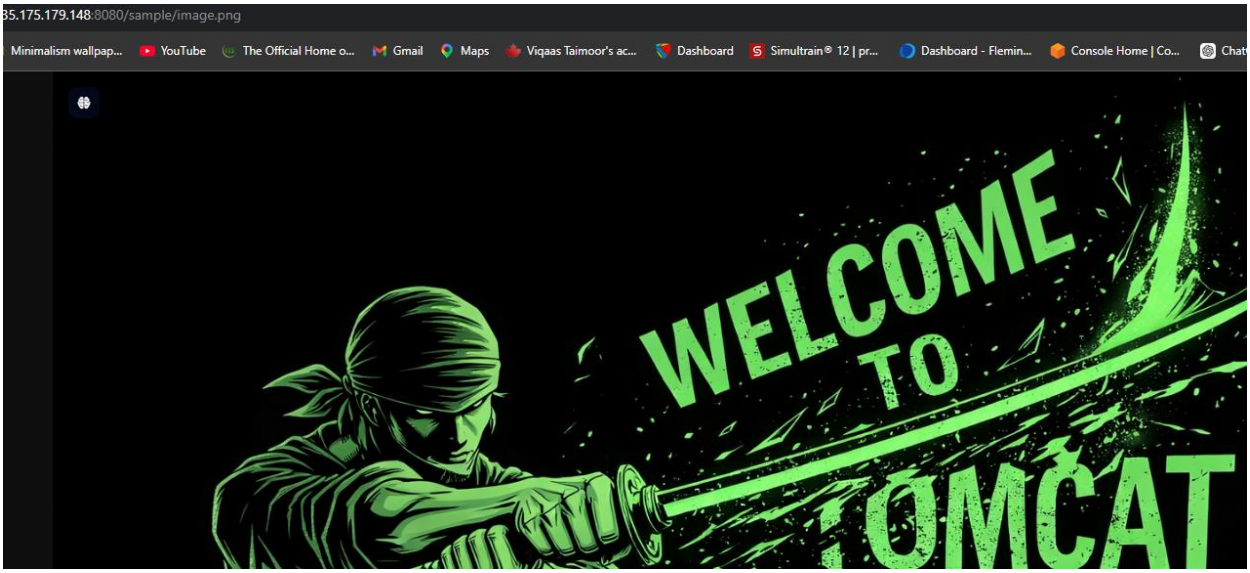
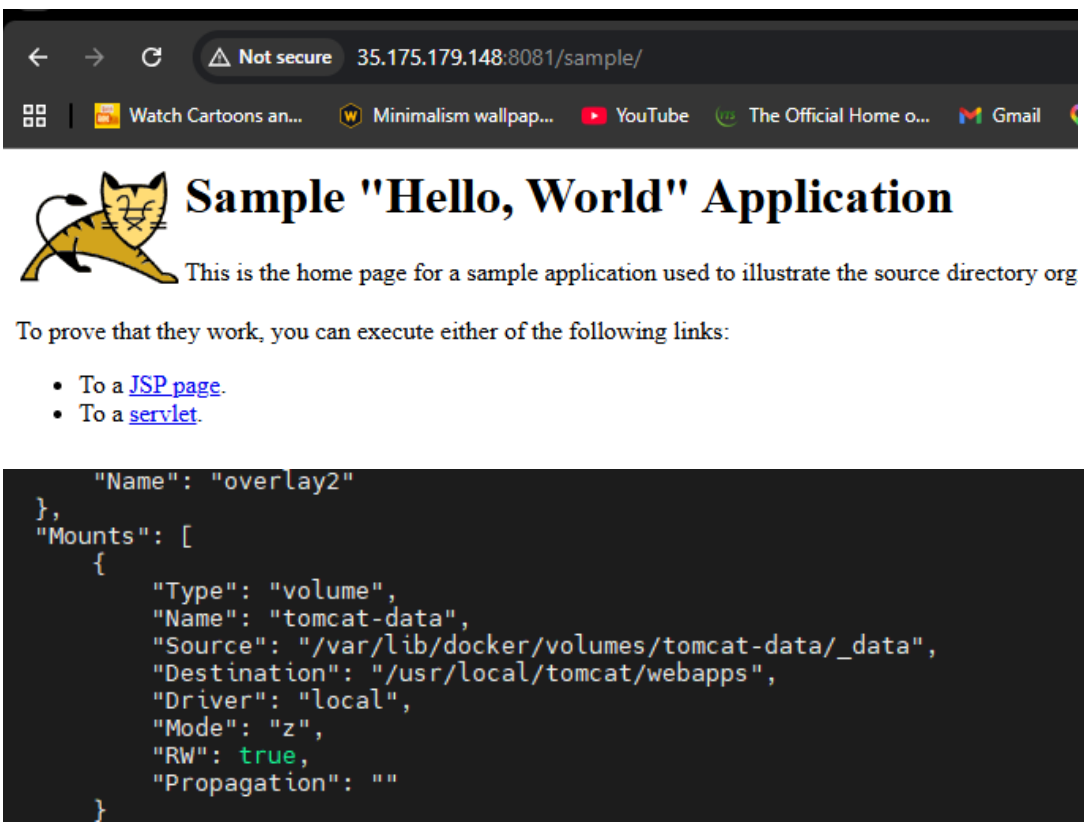
Docker Tasks - 02

- Create a tomcat container on 8080 and deploy sample application in tomcat.

```
[ec2-user@ip-172-31-24-34 ~]$ docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED         STATUS         PORTS                                                   NAMES
51cb3a568af4   tomcat            "catalina.sh run"        2 minutes ago   Up 2 minutes   0.0.0.0:8080→8080/tcp, :::8080→8080/tcp                 tomcat
a345cf68c59d   nginx             "/docker-entrypoint.…"   4 hours ago     Up 4 hours     0.0.0.0:81→80/tcp, :::81→80/tcp                         nginx
f98710bf807e   sonarqube         "/opt/sonarqube/dock…"   4 hours ago     Up 4 hours     0.0.0.0:8084→9000/tcp, :::8084→9000/tcp                 sonar
dc5689b0940e   jenkins/jenkins   "/usr/bin/tini -- /u…"   4 hours ago     Up 4 hours     50000/tcp, 0.0.0.0:8083→8080/tcp, :::8083→8080/tcp      jenkins
```



- Create volume and deploy tomcat container on port 8081.




```
        "Name": "overlay2"
    },
    "Mounts": [
        {
            "Type": "volume",
            "Name": "tomcat-data",
            "Source": "/var/lib/docker/volumes/tomcat-data/_data",
            "Destination": "/usr/local/tomcat/webapps",
            "Driver": "local",
            "Mode": "z",
            "RW": true,
            "Propagation": ""
        }
```

- Limit the nginx container to 500 MB.

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```
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT     MEM %     NET I/O         BLOCK I/O         PIDS
f77cbc769904   nginx     0.00%     4.457MiB / 500MiB     0.89%     726B / 0B       1.22MB / 12.3kB   3
b41fc72324a0   tomcat    0.04%     92.68MiB / 7.756GiB   1.17%     8.6kB / 7.45kB  0B / 827kB        35
f98710bf807e   sonar     1.00%     2.16GiB / 7.756GiB    27.85%    370kB / 2.22MB  20.5kB / 858MB    200
dc5689b0940e   jenkins   0.18%     665.5MiB / 7.756GiB   8.38%     244MB / 3.74MB  35.5MB / 683MB    48
```

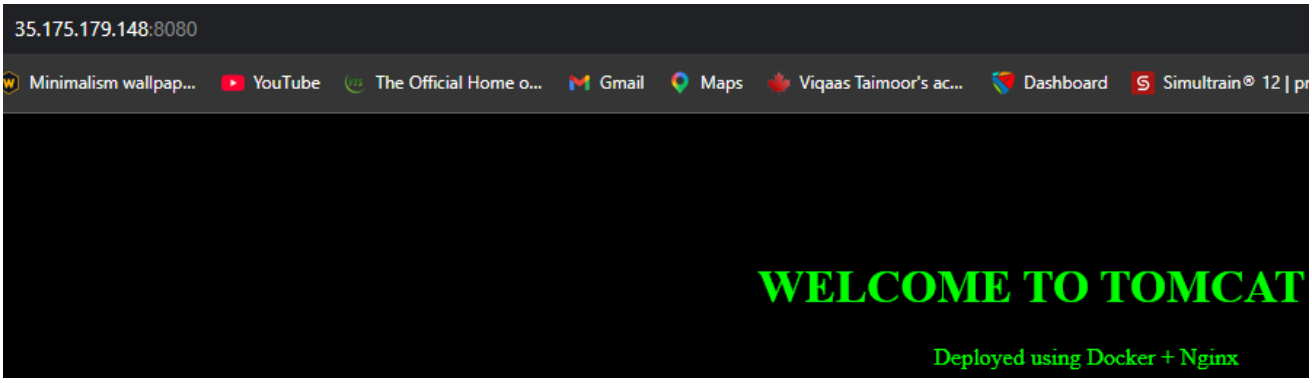
```
  "Runtime": "runc",
  "Isolation": "",
  "CpuShares": 0,
  "Memory": 524288000,
  "NanoCpus": 0,
  "CgroupParent": "",
  "BlkioWeight": 0,
  "BlkioWeightDevice": [],
  "BlkioDeviceReadBps": [],
```

- Create a sample docker file using below instructions.
    - Base module as amazonlinux:latest
    - Maintainer you name
    - Install nginx
    - COPY one index.html file to image
    - Expose on port 80
    - Command to start the nginx container

```
[root@ip-172-31-24-34 ~]# docker build -t mynginx .
[+] Building 25.7s (8/8) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 258B
 => [internal] load metadata for docker.io/library/amazonlinux:latest
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/3] FROM docker.io/library/amazonlinux:latest@sha256:162fc5b69e11e81023f83a1ab618b184927f
 => => resolve docker.io/library/amazonlinux:latest@sha256:162fc5b69e11e81023f83a1ab618b184927f
 => => sha256:162fc5b69e11e81023f83a1ab618b184927ff3002d0852432af3b6ae4a1b5304 2.38kB / 2.38kB
 => => sha256:614eae13c2d8b000b432e2ac520a63e762a56ce80a0022412fee29fe897def4b 1.02kB / 1.02kB
 => => sha256:cf7a6ca32cc6a45135eaec70c73481b2fa3452c729e2eeec0c815ed40726b4cb 583B / 583B
 => => sha256:fd916ef360dd321358d9d165369c9cdd7b0c1f945415eed426c15062374d4d1d 54.03MB / 54.03M
 => => extracting sha256:fd916ef360dd321358d9d165369c9cdd7b0c1f945415eed426c15062374d4d1d
 => [internal] load build context
 => => transferring context: 352B
 => [2/3] RUN dnf install -y nginx
 => [3/3] COPY index.html /usr/share/nginx/html/
 => exporting to image
 => => exporting layers
 => => writing image sha256:64048fae95632685c9647b0dd1c8fde7ae4256ac1e29c2a4f3aedba6a0e0bf09
 => => naming to docker.io/library/mynginx
```

Docker Tasks - 02

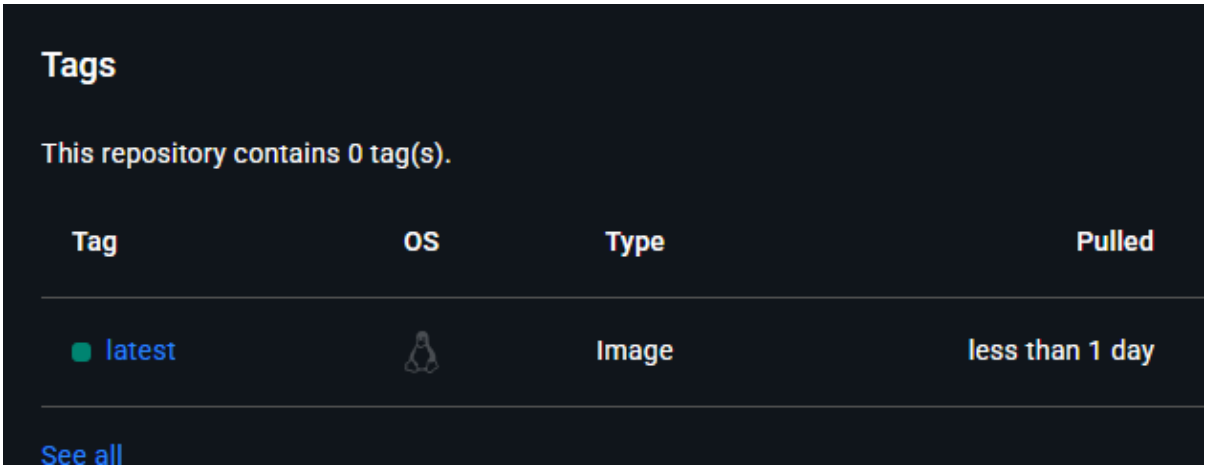
```
[root@ip-172-31-24-34 ~]# cat Dockerfile
FROM amazonlinux:latest

LABEL maintainer="V"

RUN dnf install -y nginx

COPY index.html /usr/share/nginx/html/

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```



- Push image to Docker hub

```
[root@ip-172-31-24-34 ~]# docker login
Authenticating with existing credentials ...
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
```

```
[root@ip-172-31-24-34 ~]# docker push viqaastaimoor/mynginx:latest
The push refers to repository [docker.io/viqaastaimoor/mynginx]
a50dba72c074: Pushed
18c0ad3a26ff: Pushed
4268ad241b85: Mounted from library/amazonlinux
latest: digest: sha256:34831de79ae4b5625cd3c8979ce4a2b8def073259643fa3e1e64fa759c827c27 size: 949
[root@ip-172-31-24-34 ~]#
```



- Push image to AWS ECR

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```
[root@ip-172-31-24-34 ~]# aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 526018540742.dkr.ecr.us-east-1.amazonaws.com
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[root@ip-172-31-24-34 ~]#
```

Push commands for mynginx

macOS / Linux | Windows

Make sure that you have the latest version of the AWS CLI and Docker installed. For more information, see Getting Started with Amazon ECR.

Use the following steps to authenticate and push an image to your repository. For additional registry authentication methods, including the Amazon ECR credential helper, see Registry Authentication.

1. Retrieve an authentication token and authenticate your Docker client to your registry. Use the AWS CLI:

```
aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 526018540742.d
r.ecr.us-east-1.amazonaws.com
```

Note: If you receive an error using the AWS CLI, make sure that you have the latest version of the AWS CLI and Docker installed.

2. Build your Docker image using the following command. For information on building a Docker file from scratch see th

```
[root@ip-172-31-24-34 ~]# docker tag mynginx:latest 526018540742.dkr.ecr.us-east-1.amazonaws.com/mynginx:latest
[root@ip-172-31-24-34 ~]# docker push 526018540742.dkr.ecr.us-east-1.amazonaws.com/mynginx:latest
The push refers to repository [526018540742.dkr.ecr.us-east-1.amazonaws.com/mynginx]
a50dba72c074: Pushed
18c0ad3a26ff: Pushed
4268ad241b85: Pushed
latest: digest: sha256:34831de79ae4b5625cd3c8979ce4a2b8def073259643fa3e1e64fa759c827c27 size: 949
[root@ip-172-31-24-34 ~]#
```

Images (1) Info

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latest	Image	March 24, 2026, 17:04:13 (UTC+05.5)	179.11